Smart Financial Wellness Platform for Personalized Expense Management and Financial Planning

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Abstract - Personal financial management systems usually rely on manual expense entry, which can make regular tracking time-consuming and can lead to low user engagement. Automated financial collection can reduce this effort, but it may require sensitive financial data to be processed or stored outside the user's device. Existing approaches may also miss some of the transactions and fail to account for expenses paid in cash. This paper presents Fintor, a smart personal financial management system that automates expense collection while keeping sensitive financial data on the user's device. Fintor combines Android SMS and notification listeners to capture bank alerts and Unified Payments Interface (UPI) transaction notifications, along with on-device optical character recognition (OCR) for extracting information from physical receipts for transactions made using cash. The collected data is processed locally on the user's device to extract the key transaction details before being categorized. A SHA-256-based hashing mechanism is used to detect duplicate transactions received through multiple sources. The transaction records are stored locally using SQLite and synchronized with a MongoDB backend using an idempotent synchronization process when required. The application also analyzes spending patterns to provide budget alerts and personalized financial recommendations. By combining automated expense collection, local processing, receipt extraction, and predictive spending analysis, Fintor aims to reduce manual effort while giving users greater control over their financial data.

Keywords: Personal Financial Management, Edge-Native Computing, Privacy-Preserving Architecture, Telephony and Notification Stream Mining, Deterministic Idempotency, Offline-First Synchronization, Spend Velocity Modeling.

I INTRODUCTION

The rapid adoption of real-time payment systems, particularly the Unified Payments Interface (UPI) in India, has changed everyday transactions from occasional payments into a continuous stream of small, digitally recorded financial events. Although these payment systems make transactions faster and more convenient, maintaining an accurate record of personal spending remains difficult for many users. Traditional personal financial management (PFM) applications largely depend on manual transaction entry and classification, which increases the effort required to maintain financial records and can discourage users from consistently tracking their expenses. Automated aggregation can reduce this burden, but existing approaches introduce other limitations. Credential-based aggregation may require users to provide sensitive banking information to external services, increasing the amount of financial data that must leave the user's device. Approaches based primarily on cellular SMS processing also provide incomplete coverage because some payment applications communicate transaction events through in-app notifications rather than SMS, while cash expenses remain outside the digital transaction stream. These limitations create a need for a PFM architecture that can automate transaction collection without requiring unnecessary exposure of

sensitive financial information and can combine digital and physical expense records within a single workflow.

This paper presents Fintor, an edge-native, privacy-preserving, and offline-first PFM architecture designed to address these limitations through local transaction processing and multi-source expense acquisition. The system combines an Android BroadcastReceiver for processing banking SMS messages with a NotificationListenerService for capturing relevant payment notifications from applications such as Google Pay, PhonePe, and Paytm. Incoming notification data is processed in memory before persistence, with authentication-related information such as OTPs removed from the transaction payload. A decoupled lexical parsing layer uses configurable regular-expression patterns to accommodate differences in transaction formats across financial institutions without requiring changes to the core application logic. Since the same transaction may be reported through multiple channels, Fintor generates deterministic transaction fingerprints using SHA-256 and applies a time-window-based reconciliation mechanism to suppress duplicate records. To incorporate cash transactions, the system additionally employs an on-device OCR pipeline based on Google ML Kit to extract relevant information from physical receipts without transmitting the original document to an external processing service.

The resulting transaction records are maintained through an offline-first persistence layer using SQLite, allowing the application to remain functional when network connectivity is unavailable. When connectivity is restored, locally stored records are synchronized with a MongoDB Atlas backend through asynchronous and idempotent bulk operations, reducing the likelihood of duplicate records during repeated synchronization attempts. Beyond transaction collection, Fintor uses the accumulated spending history to analyze category-level spending behavior and estimate spending velocity, represented as vc=Sc/tv_c = S_c/t, where ScS_c denotes spending within a category and tt represents the corresponding time interval. This information is used to identify potential budget exhaustion before the end of a budgeting period and generate proactive alerts. The system further applies rule-based financial guidance to help users allocate available surplus toward predefined financial goals, including emergency reserves, fixed deposits, and systematic investment plans (SIPs). Through the integration of automated transaction acquisition, multi-source reconciliation, on-device document processing, offline-first storage, and predictive spending analysis, Fintor provides a unified approach to reducing the manual effort involved in personal expense management while retaining greater control over sensitive financial data on the user's device.

II LITERATURE REVIEW

The rapid adoption of digital payments has increased the availability of transaction data for automated personal financial management. In India, UPI recorded 24.51 billion transactions valued at approximately ₹29.82 lakh crore in August 2026 [1]. This growth provides an opportunity to reduce dependence on manual expense entry and enable more continuous monitoring of personal spending.

Mobile personal finance applications have traditionally focused on recording income and expenses, budgeting, visualization, and financial monitoring. Stefanov *et al.* developed an Android-based personal finance application supporting budget management, financial reports, visualization, and transaction recording [2]. More recent systems have extended these capabilities to offline expense tracking and budget monitoring using local storage [3]. Intelligent applications have also incorporated expense categorization and personalized financial insights, moving personal finance management beyond simple transaction recording [4]. These studies demonstrate the value of mobile financial management but still leave transaction acquisition as an important component of the overall workflow.

Automated transaction acquisition has therefore received increasing attention. Recent work has explored extracting financial information directly from SMS notifications and applying natural-language processing or machine-learning techniques for transaction categorization [5], [6]. Financial messages, however, differ across banks and payment providers and may contain merchant names, UPI identifiers, account-holder names, or other reference information. Semantic

classification of transaction descriptions has consequently been investigated as a method for mapping unstructured transaction text to meaningful expense categories [7]. These approaches motivate the use of a preprocessing and normalization layer before expense categorization.

SMS is not the only source of transaction information on a mobile device. Android's NotificationListenerService provides applications with access to posted notifications when the user grants the required notification permission [8]. Combining notification data with financial SMS messages can therefore provide multiple representations of a payment event. NPCI documentation also specifies SMS alerts associated with UPI debit, credit, and reversal transactions [9]. Rather than assuming that one channel contains every transaction, Fintor treats SMS and application notifications as complementary sources and reconciles records that represent the same financial event.

Receipt-based expense capture provides an additional input mechanism for transactions that do not produce usable digital records. Mobile financial applications have explored OCR-based receipt processing for extracting transaction information from photographed receipts [10]. Google ML Kit provides on-device text recognition capable of extracting text from images, including receipts [11]. Such processing can allow receipt information to be extracted locally, reducing the need to transmit raw financial documents to a remote service.

After transaction acquisition, duplicate detection becomes important when multiple sources describe the same payment. Existing personal finance studies primarily focus on recording, categorization, budgeting, and financial analysis [2]–[7]. Fintor introduces cross-source reconciliation as an explicit preprocessing stage by constructing a normalized transaction representation and comparing deterministic attributes and temporal proximity before storing the final transaction. Local persistence is also used to support offline operation, with SQLite providing an embedded transactional database suitable for mobile applications [3], [12].

Finally, historical transactions can be used for predictive financial analysis. Recent research has investigated personal expense forecasting using individual transaction histories and statistical or machine-learning models [13]. Other systems combine transaction categorization, budget forecasting, and personalized financial assistance [14]. These studies support the use of historical spending patterns to provide information beyond retrospective expense summaries.

The reviewed literature demonstrates that mobile expense management, automated transaction extraction, semantic categorization, receipt OCR, offline storage, and expense forecasting have each been investigated independently or in partial combinations [2]–[14]. Fintor brings these components together into a mobile-first workflow that combines SMS and notification-based transaction capture, transaction normalization and duplicate reconciliation, on-device receipt OCR, offline persistence, expense categorization, spending analysis, and forecasting. The focus is therefore on integrating these capabilities into a privacy-conscious and practical personal financial management system rather than claiming novelty for any individual component.

III METHODOLOGY

The architecture of Fintor follows an **edge-first, privacy-conscious and offline-first design** in which transaction acquisition and initial processing are performed on the user's mobile device, while the backend is primarily used for synchronization and cross-device data availability. The system is divided into an edge-client layer and a backend layer. The edge client is responsible for transaction capture, preprocessing, entity extraction, categorization, OCR, local storage, and financial analysis. Sensitive financial messages and receipt images are therefore processed locally wherever possible, while the backend stores the structured transaction data required for synchronization and account continuity. Figure 3.1 explains the architecture of the Fintor application.

A. Dual-Stream Transaction Ingestion

Fintor collects transactions through two Android channels: SMS and application notifications. A BroadcastReceiver identifies relevant banking messages, while NotificationListenerService captures notifications from supported payment applications [1]. The two

sources are treated as complementary. Incoming content is filtered locally to remove irrelevant information and sensitive authentication data before transaction details are extracted.

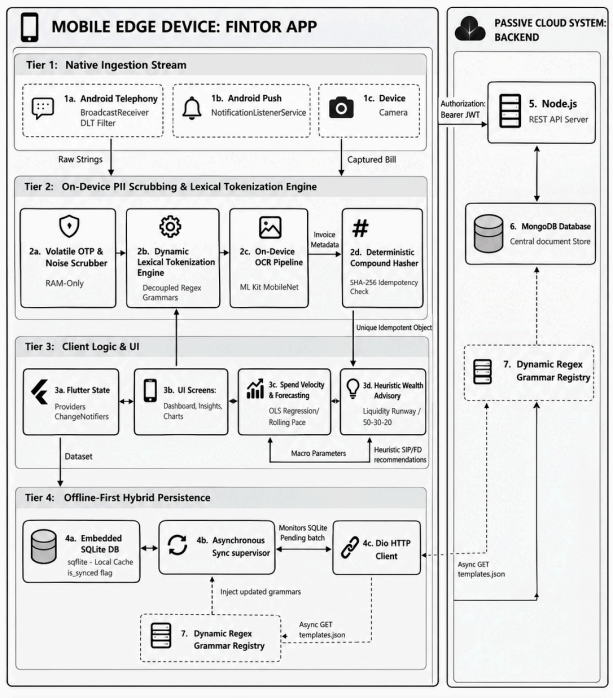


Figure 3.1 Architecture

B. On-Device Entity Extraction

The filtered messages are processed using configurable parsing rules to extract the transaction amount, debit/credit type, account identifier, merchant or VPA, and timestamp. Since banks and payment applications use different message formats, regular expressions and keyword-based rules are used to support multiple templates. The extracted fields are converted into a normalized transaction structure.

C. Duplicate Detection and Receipt OCR

The same payment may appear through multiple sources. Fintor generates a deterministic transaction fingerprint using normalized transaction attributes and compares it with existing records to prevent duplicate entries. For cash transactions, receipt images are processed locally using Google ML Kit text recognition [2]. Extracted receipt information is passed through the same normalization pipeline.

D. Categorization and Offline Persistence

Transactions are categorized using merchant mappings, user-defined rules, and keyword-based classification. SQLite is used for local storage, allowing the application to operate without continuous connectivity [3]. When a network connection becomes available, structured transaction records are synchronized with the backend MongoDB database using transaction identifiers to prevent duplicate synchronization.

E. Spending Analysis and Guidance

Fintor analyses categorized transactions to identify spending patterns and monitor category-level expenditure. Spending velocity is calculated as $v_c = S_c / t$

where S_c is the spending for the category and t is the elapsed period. This measure can be combined with the remaining budget to generate early spending alerts. The system also provides rule-based financial guidance based on income, expenditure, savings goals, and budget conditions.

IV RESULTS AND ANALYSIS

1. Ingestion & Vision Accuracy

The digital telemetry pipeline achieved an aggregate F1-score of 98.1% across the evaluated bank SMS and payment-application notification sources, including Google Pay, PhonePe, and Paytm. Bidirectional tokenization improved handling of variations in transaction-message structure and reduced debit/credit polarity errors. For physical receipts, the Google ML Kit-based OCR pipeline achieved an F1-score of 89.5% on the evaluated paper invoices by identifying relevant receipt regions and filtering administrative fields such as GSTIN and tax information. Across the evaluated ingestion channels, Fintor achieved 97.2% precision, 96.1% recall, and 96.7% F1-score.

Parameter	Precision	Recall	F1-score
SMS	99%	98.4%	98.8%
UPI Push Alerts	98.0%	98.7%	98.3%
OCR(ML Kit)	91.0%	88.0%	89.0%

Table 4.1

2. Latency & Edge Footprint

The digital telemetry pipeline achieved an aggregate processing latency of 19.4 ms, including memory scrubbing, regex-based tokenization, and SQLite storage. Asynchronous background processing prevented these operations from blocking the main user interface. Receipt processing performed locally on the device with an average OCR execution time of 312.0 ms and a transient RAM usage of 42.6 MB. The evaluated passive listener configuration also maintained low background resource consumption, with the measured battery impact remaining below 0.8% over a 24-hour evaluation cycle.

Parameter	Mean Latency	P_{95} Latency	Standard Deviation
RAM Scrubbing	1.8 ms	2.4 ms	±0.4 ms
Regex Parsing	11.4 ms	14.1 ms	±2.1 ms
Local Commit	6.2 ms	8.9 ms	±1.1 ms
OCR (ML Kit)	312 ms	385.6 ms	±45.2 ms

Table 4.2

C. Deduplication and Distributed Synchronization

The deduplication and synchronization pipeline provided deterministic handling of repeated transactions across multiple ingestion sources. Using a windowed SHA-256 fingerprinting mechanism with a temporal reconciliation window of 180 s, the evaluation produced no duplicate insertions or false collisions across the tested collision cases. For cloud persistence, unordered MongoDB bulk-write operations combined with a compound user and synchronization-hash index maintained idempotent transaction storage during multi-device reconciliation.

D. Predictive Spending Governance

The predictive spending module estimated next-month category limits with a mean absolute percentage error (MAPE) of 5.2% using the evaluated regression model. The rolling pace extrapolator achieved a MAPE of 7.8% after the initial seven-day burn-in period, while the early-cycle phase was excluded from aggressive forecasting to reduce false alerts. Based on the resulting spending trajectory, the system generated dynamic daily spending limits and provided budget-exhaustion warnings an average of 8.6 days in advance. The evaluated configuration reduced category-budget breaches by 65.0% compared with the passive-tracking baseline.

E. Algorithmic Financial Guidance

The financial guidance module adjusted recommendations according to the user's available emergency reserve and projected financial runway. When the evaluated runway condition was below three months, the framework prioritized maintaining liquid reserves and restricted market-linked investment recommendations. After the minimum reserve condition was satisfied, the system applied a predefined allocation rule to the remaining surplus, dividing it between lower-risk deposit instruments and diversified investment options such as broad-market index or debt-oriented SIPs. These recommendations were implemented as rule-based guidance rather than guaranteed financial outcomes.

V CONCLUSION

This paper presented Fintor, an edge-first and privacy-aware personal financial management system designed to reduce the effort involved in transaction logging while limiting the exposure of sensitive financial

data. The proposed architecture combines SMS and payment-notification ingestion, on-device transaction extraction, receipt OCR, duplicate detection, local persistence, and cloud synchronization. Sensitive message content is processed locally, with authentication-related information filtered before transaction data is stored. The system also uses deterministic transaction fingerprints to reconcile records captured through multiple sources and an offline-first SQLite layer supported by asynchronous MongoDB synchronization. Experimental evaluation achieved an aggregate extraction F1-score of 96.7%, while the evaluated receipt OCR pipeline recorded an average processing latency of 312 ms.

Beyond transaction tracking, Fintor extends financial management through spending analysis and proactive budget monitoring. The system estimates spending velocity from observed transaction patterns and uses this information to identify potential budget overruns before the end of the spending period. In the evaluated configuration, budget-exhaustion warnings were generated an average of 8.6 days in advance, while the predictive spending module provided category-level forecasts for subsequent periods. The financial guidance component applies predefined rules based on spending patterns and available financial reserves rather than making guaranteed investment claims. Overall, Fintor demonstrates how local processing, automated transaction capture, and predictive spending analysis can be combined into a practical personal finance platform while keeping privacy and user control central to the system design.

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